

The Search for Extrasolar Planets

Since it appears the conditions for planet formation are common, we'd like to know how many solar systems there are, and what they look like.

Indirect Methods:

- 1) Doppler shift of the star's orbit
this is the main one so far
- 2) Astrometric wobble of the star's orbit

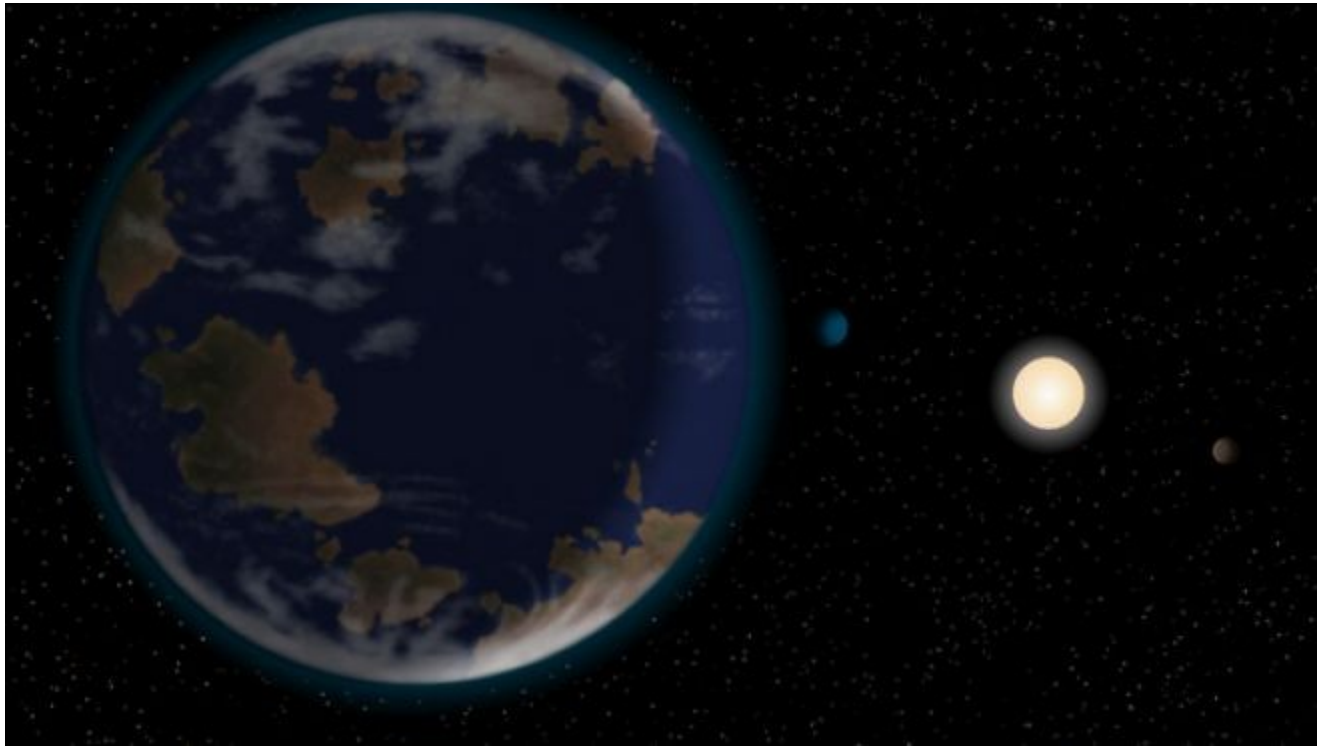
Semi-direct Methods:

- 1) Transits (light blocked by the planet)
might also see phases
- 2) Microlensing (planet's gravity)

Direct Methods:

- 1) Planet imaged directly (perhaps with coronagraph)
reflected or emitted (IR or radio) light
- 2) Planet imaged by interferometer





This artist's impression shows the newfound potentially habitable alien planet HD40307g in the foreground, with its host star and two other worlds in the six-planet system also depicted. The atmosphere and continents shown are neither detected nor confirmed (J. Pinfield, for the RoPACS network at the University of Hertfordshire.)

Read more: <http://www.foxnews.com/science/2012/11/07/upper-earth-alien-planet-may-be-habitable-for-life/#ixzz2BeaxSUMs>

Detecting Extrasolar Planets

Our goals for learning:

- Why is it so difficult to detect planets around other stars?
- How do we detect planets around other stars?

first extrasolar planet : 1992 , 3 object EArth like masses orbiting a pulsar (from a supernova debris)

1995: first planet orbiting a star called 51 Pegasi . by Astronomers Michel Mayor and Didier Queloz confirmed by Geoffrey W. Marcy and Paul Butter of San Francisco University.

<http://news.discovery.com/videos/space-3-questions-exoplanets.html>

See video from discovery channel – aliens and see other links.

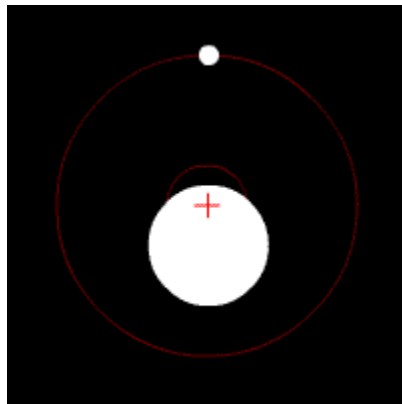
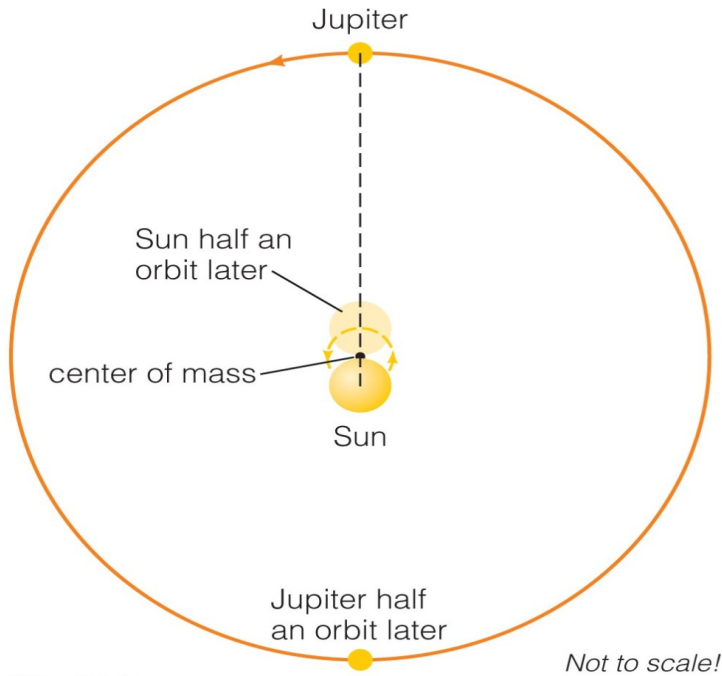
Brightness Difference

- A Sun-like star is about a billion times brighter than the light reflected from its planets.
- This is like being in San Francisco and trying to see a pinhead 15 meters from a grapefruit in Washington, D.C.

Planet Detection

- **Direct:** pictures or spectra of the planets themselves
- **Indirect:** measurements of stellar properties revealing the effects of orbiting planets

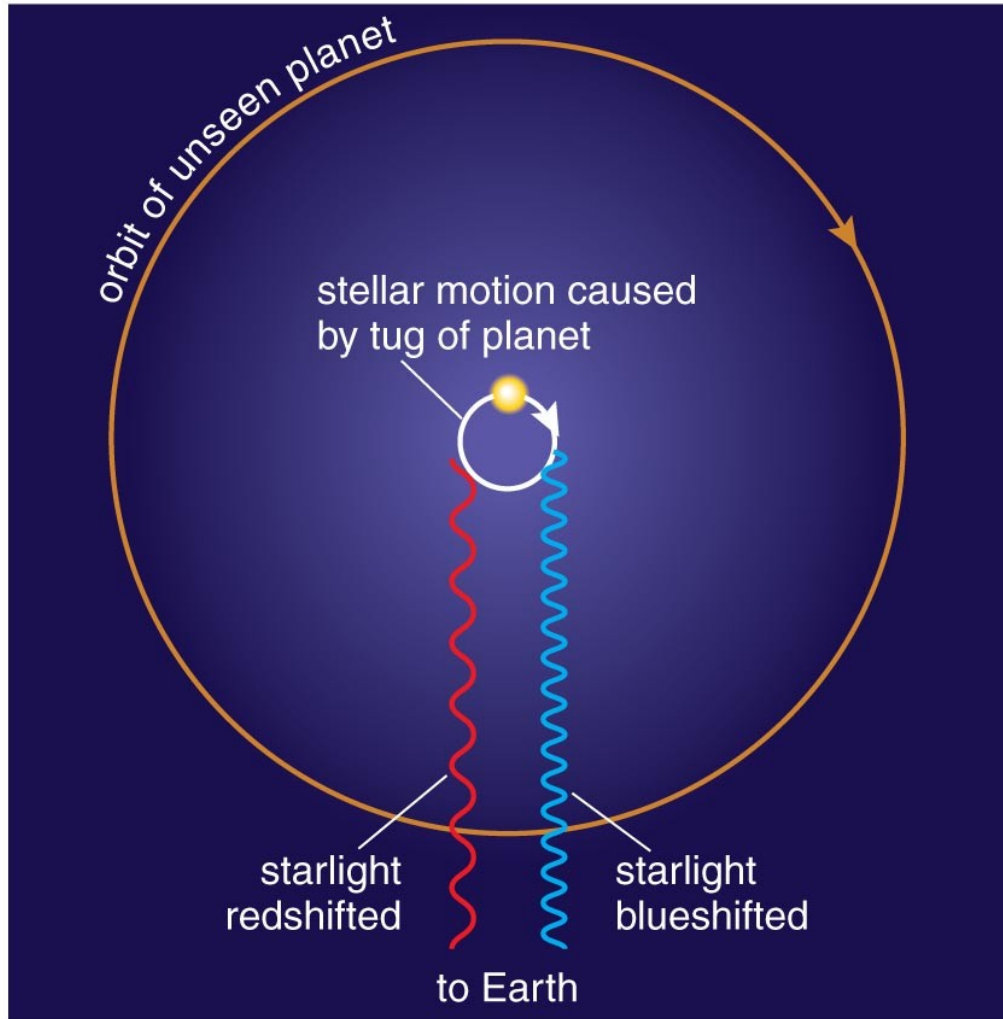
Gravitational Tugs



- The Sun and Jupiter orbit around their common center of mass.
- The Sun therefore wobbles around that center of mass with same period as Jupiter.

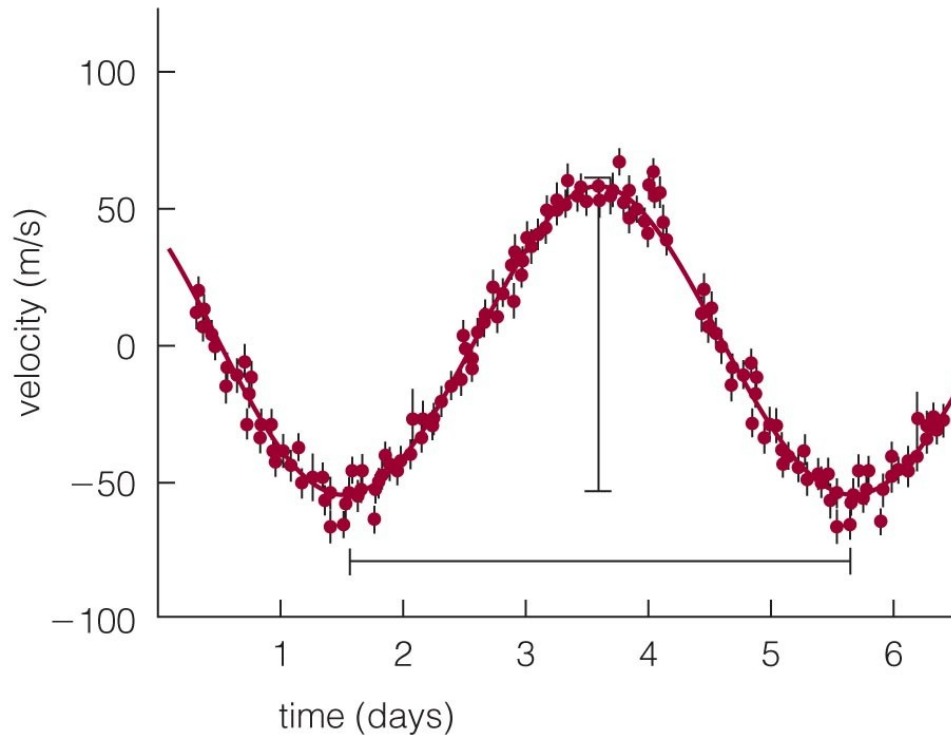
See lecture 38. understanding the universe.
15:30

Doppler Technique



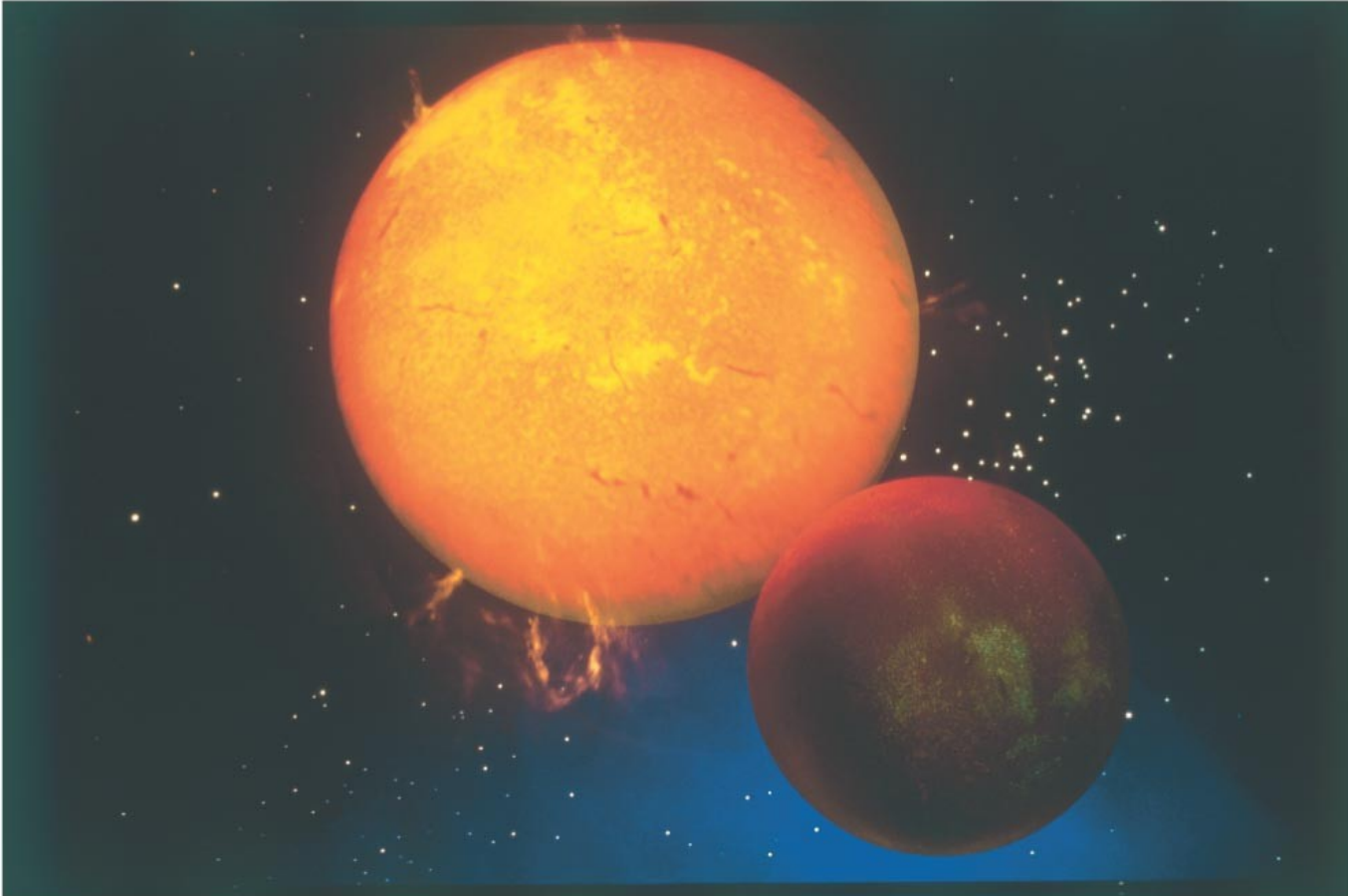
- Measuring a star's Doppler shift can tell us its motion toward and away from us.
- Current techniques can measure motions as small as 1 m/s (walking speed!).

First Extrasolar Planet



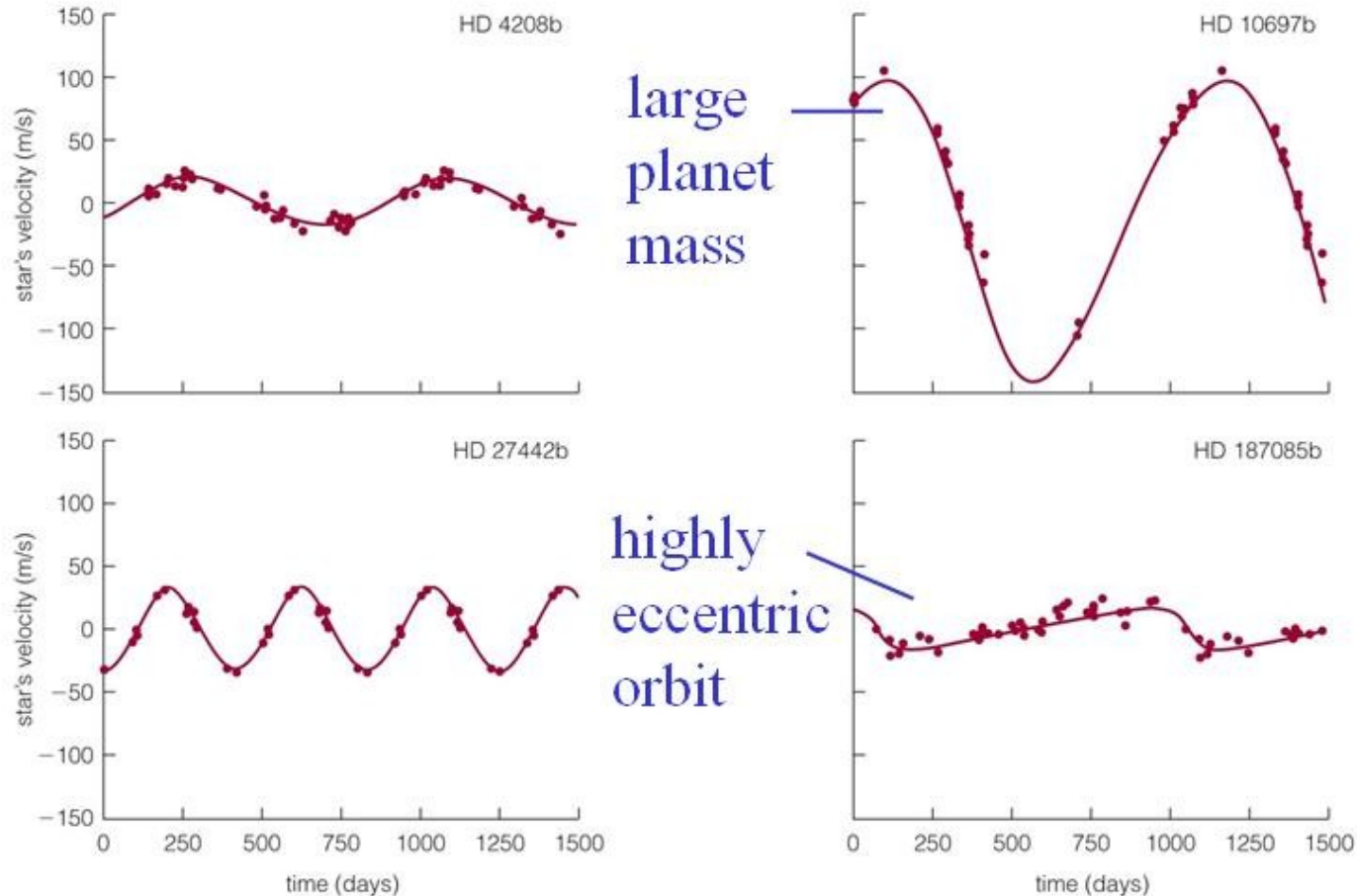
- Doppler shifts of the star 51 Pegasi indirectly revealed a planet with 4-day orbital period.
- This short period means that the planet has a small orbital distance.
- This was the first extrasolar planet to be discovered (1995).

First Extrasolar Planet



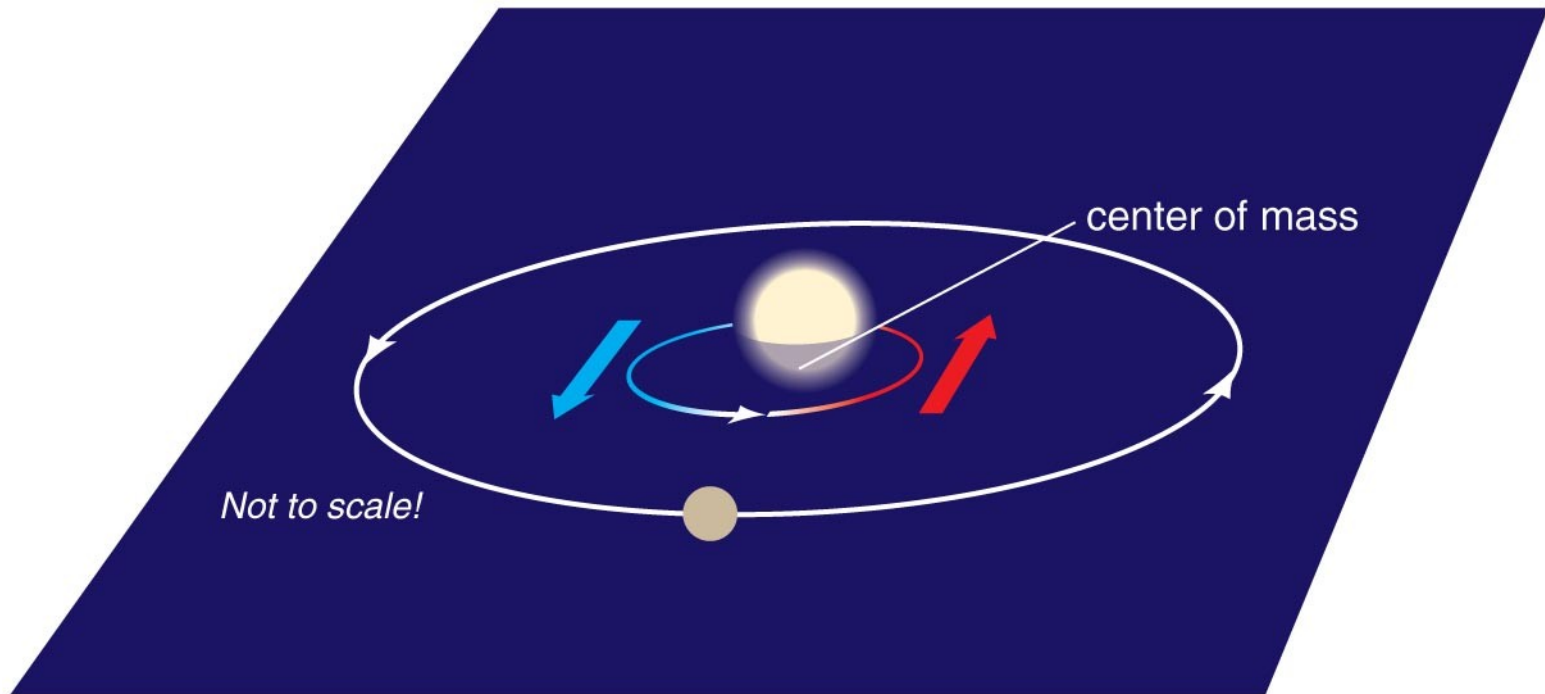
- The planet around 51 Pegasi has a mass similar to Jupiter's, despite its small orbital distance.

Other Extrasolar Planets



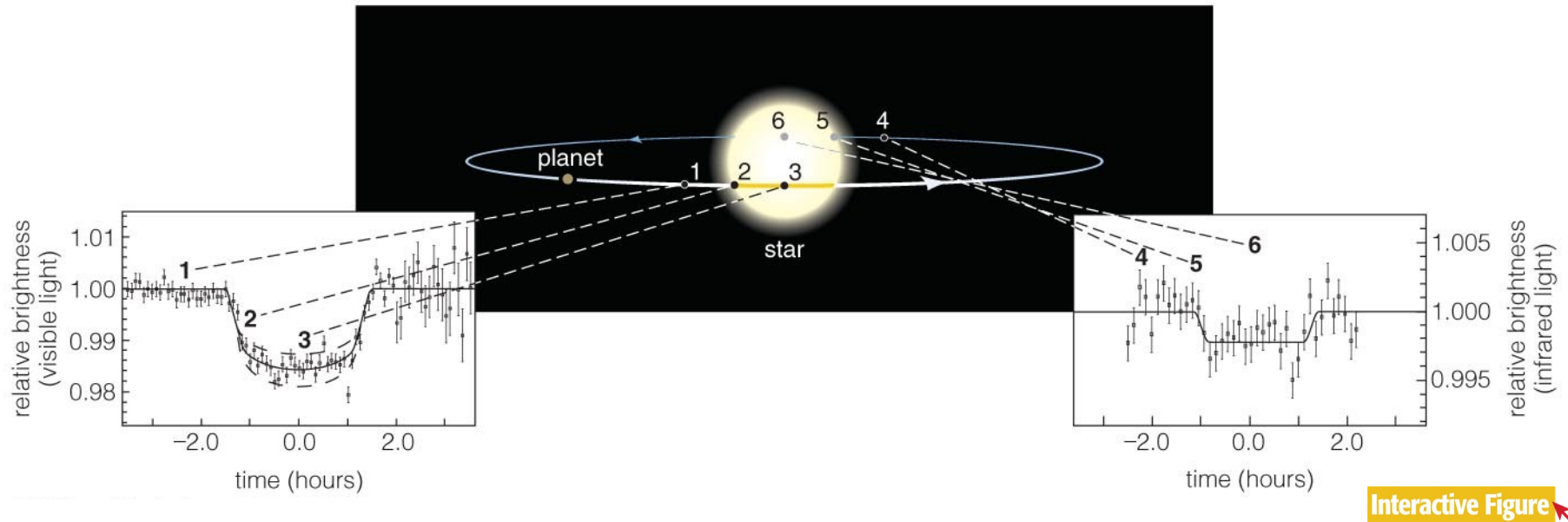
- Doppler shift data tell us about a planet's mass and the shape of its orbit.

Planet Mass and Orbit Tilt



- We cannot measure an exact mass for a planet without knowing the tilt of its orbit, because Doppler shift tells us only the velocity toward or away from us.
- Doppler data give us lower limits on masses.

Transits and Eclipses

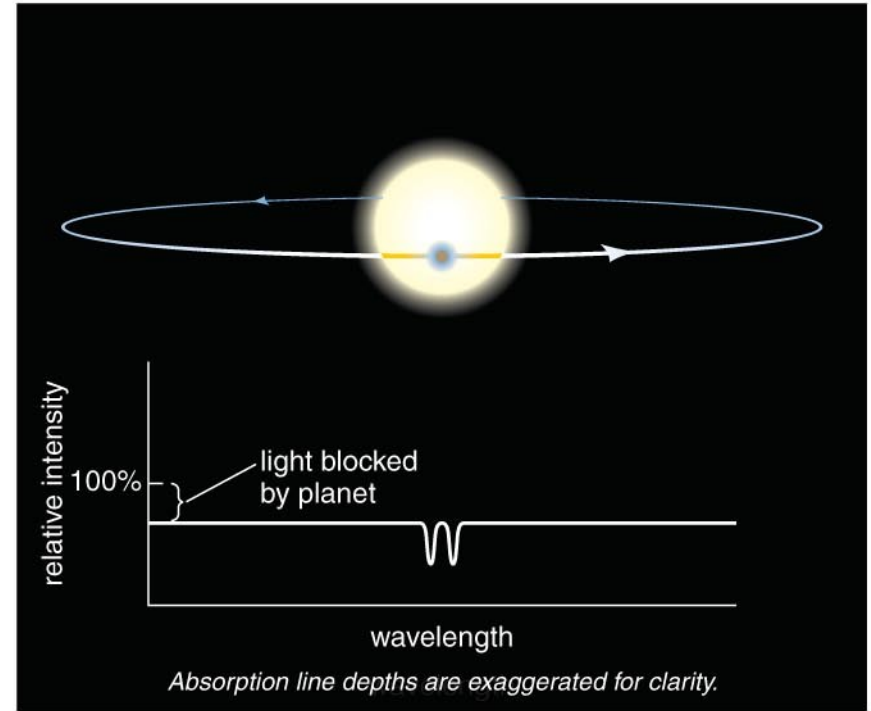
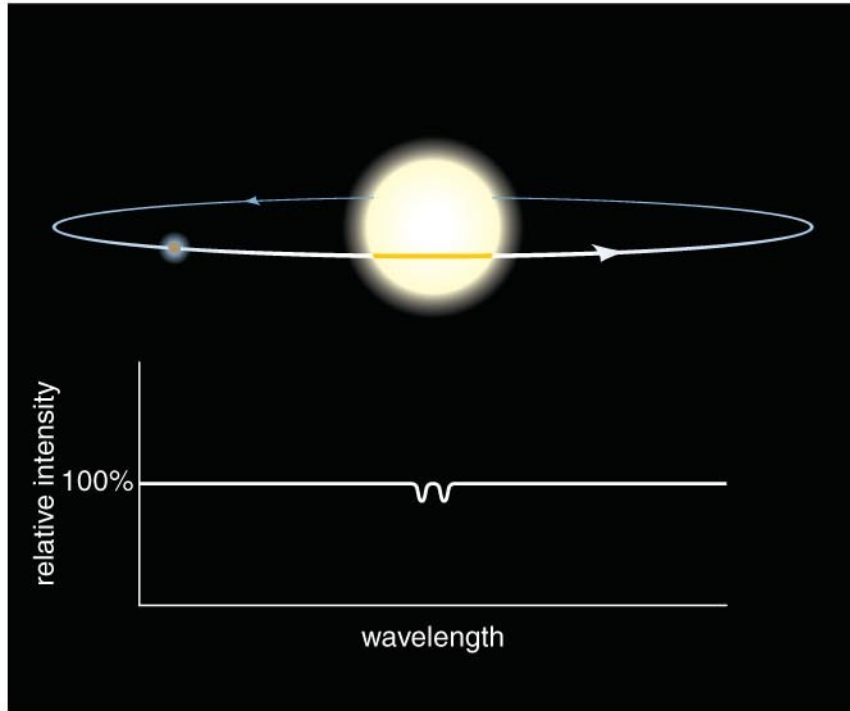


Interactive Figure

- A **transit** is when a planet crosses in front of a star.
- The resulting eclipse reduces the star's apparent brightness and tells us planet's radius.
- No orbital tilt: accurate measurement of planet mass

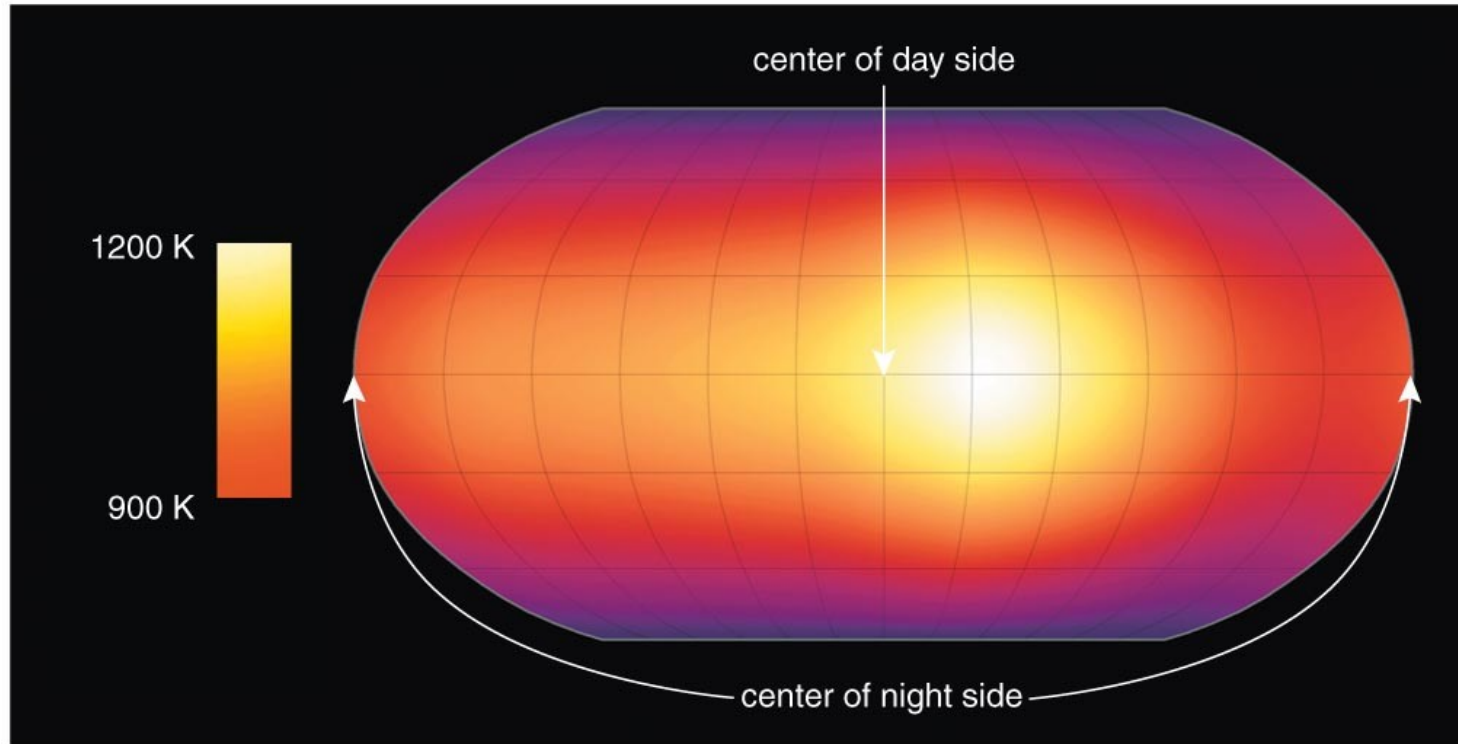
Lecture 39:extra solar planets galore.
20:45

Spectrum During Transit



- Change in spectrum during a transit tells us about the composition of planet's atmosphere.

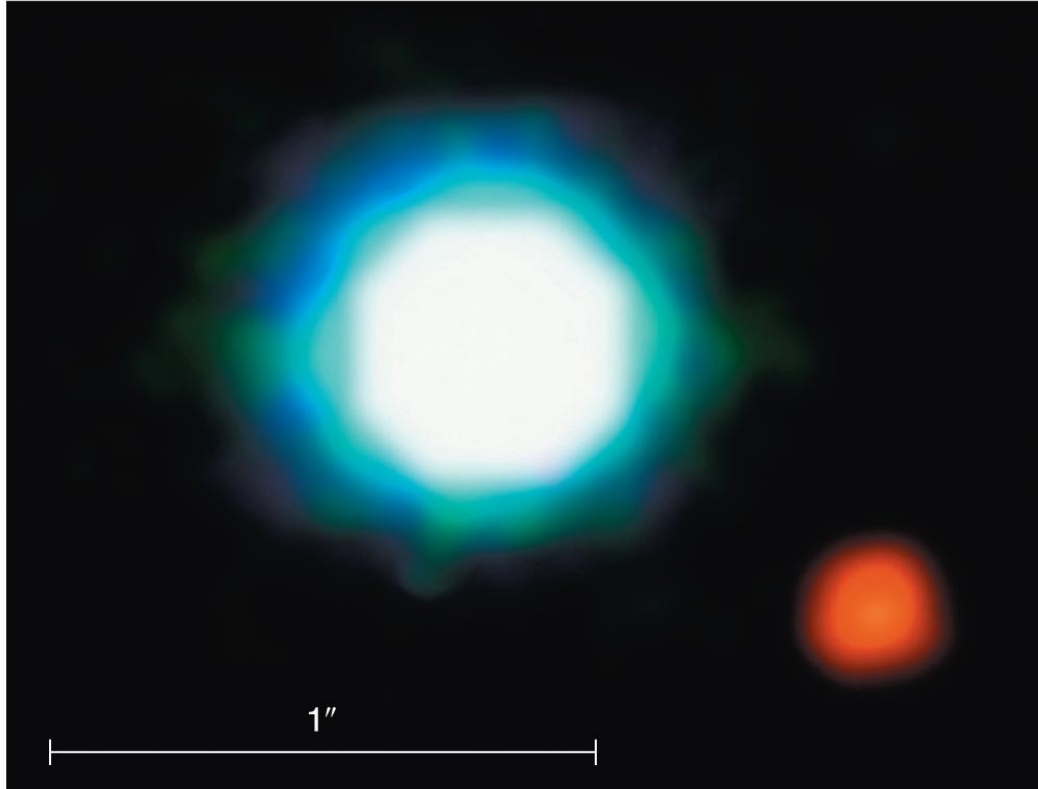
Surface Temperature Map



- Measuring the change in infrared brightness during an eclipse enables us to map a planet's surface temperature.

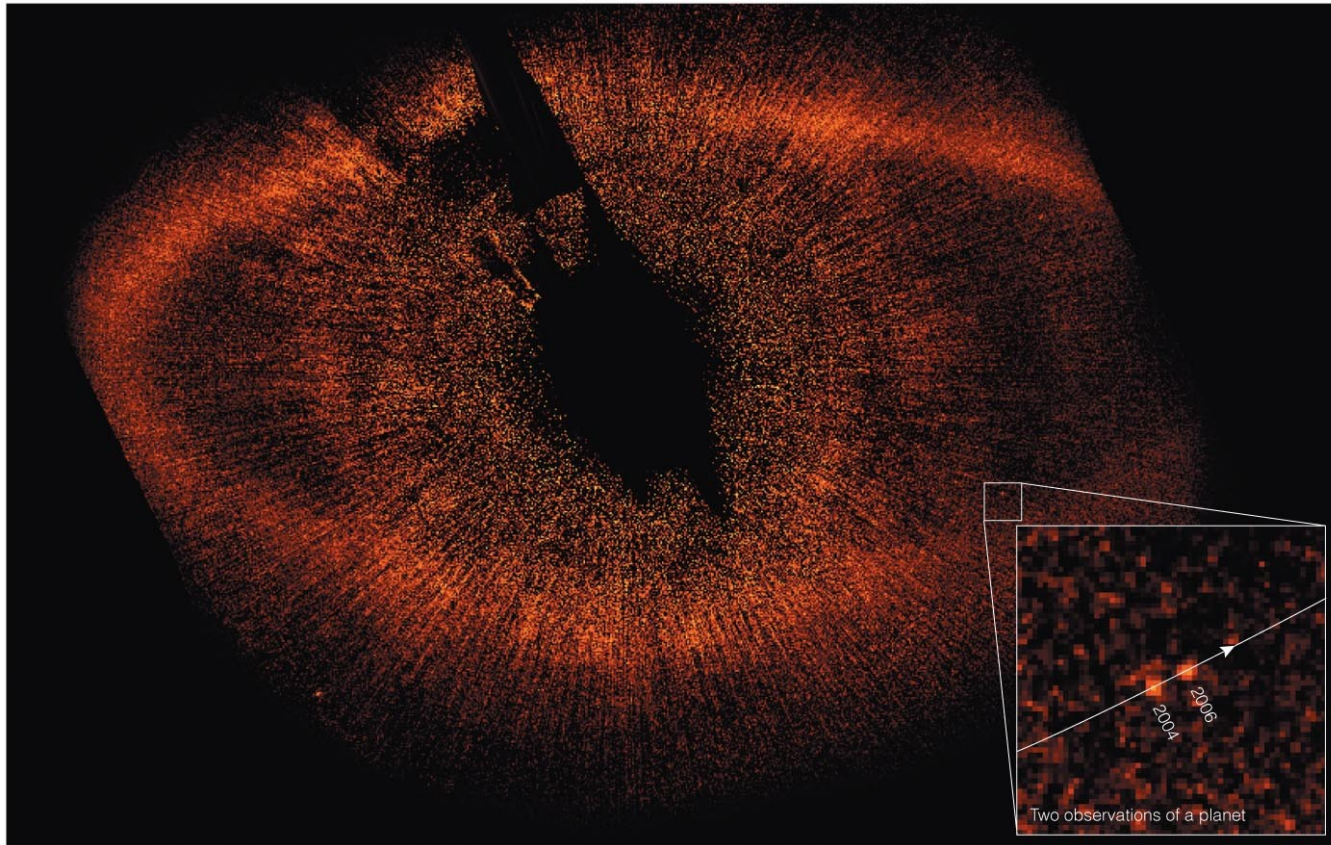
Direct Detection

- Special techniques like adaptive optics are helping to enable direct planet detection.



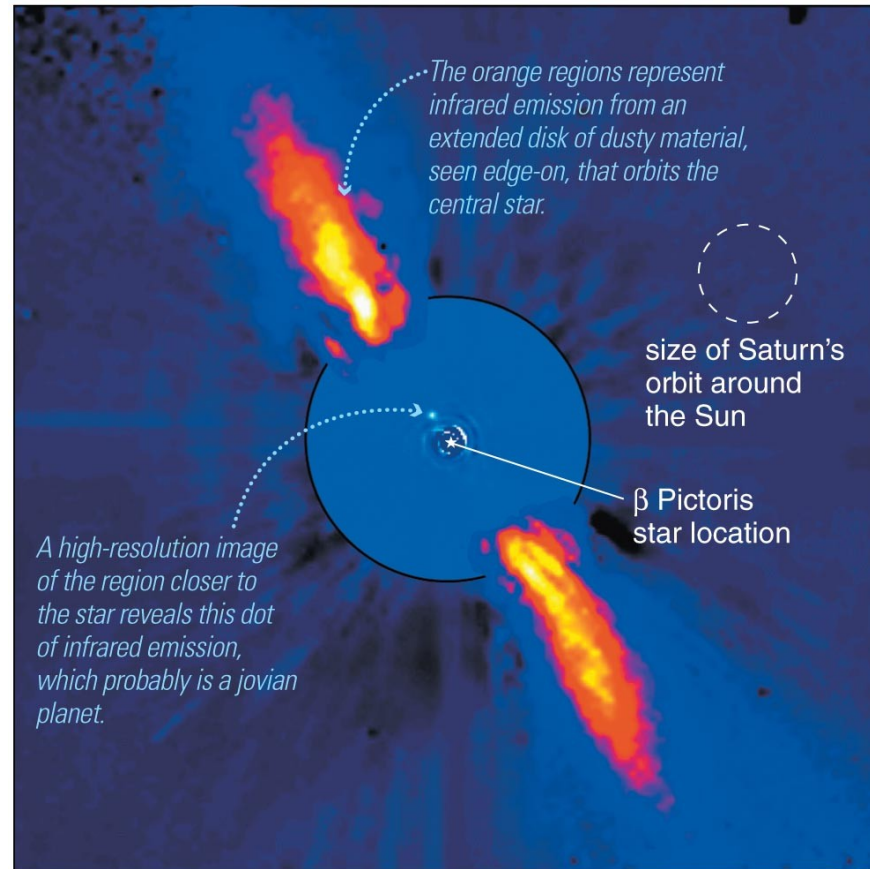
The first ever image of a planet outside our Solar System: the European Southern Observatory's VLT (Very Large Telescope) captured this image of a Brown Dwarf star (center) and its planetary companion. World's most advanced optical instrument. The VLT is located at Paranal Observatory in the Atacama Desert.

Direct Detection



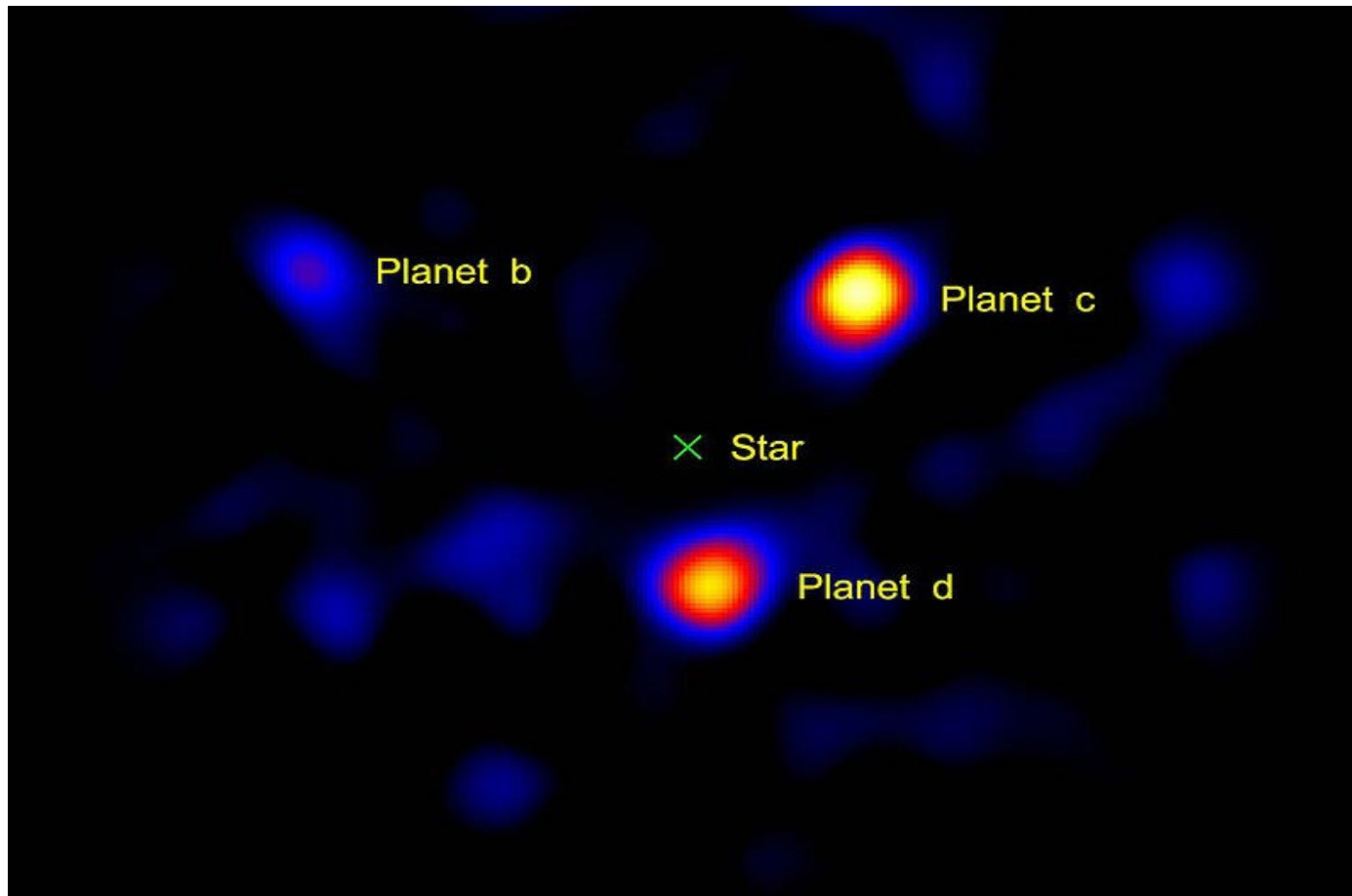
- Techniques that help block the bright light from stars are also helping us to find planets around them.

Direct Detection



- Techniques that help block the bright light from stars are also helping us to find planets around them.

The new exoplanet, Beta Pictoris b



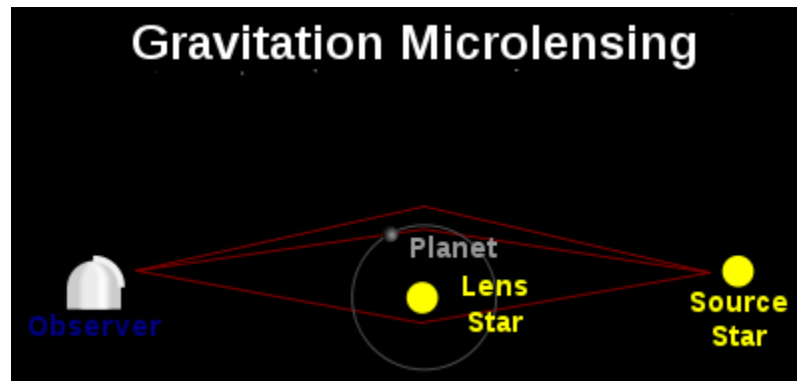
This image shows the light from three planets orbiting a star 120 light-years away. The planets' star, called HR8799, is located at the spot marked with an "X."

This picture was taken using a small, 1.5-meter (4.9-foot) portion of the Palomar Observatory's Hale Telescope, north of San Diego, Calif. This is the first time a picture of planets beyond our solar system has been captured using a telescope with a modest-sized mirror -- previous images were taken using larger telescopes.

The three planets, called HR8799b, c and d, are thought to be gas giants like Jupiter, but more massive. They orbit their host star at roughly 24, 38 and 68 times the distance between our Earth and sun, respectively (Jupiter resides at about 5 times the Earth-sun distance).

Other Planet-Hunting Strategies

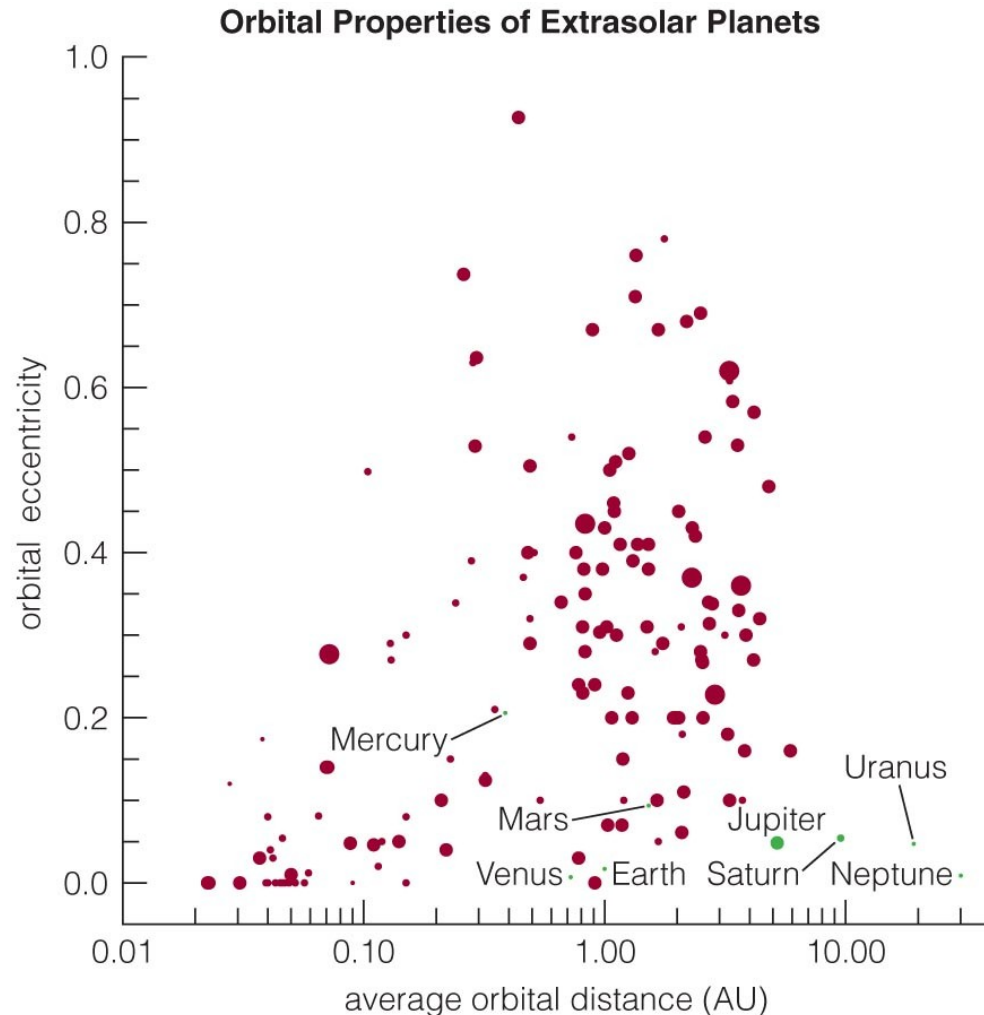
- **Gravitational Lensing:** Mass bends light in a special way when a star with planets passes in front of another star.



29:10
Lecture 39

- **Features in Dust Disks:** Gaps, waves, or ripples in disks of dusty gas around stars can indicate presence of planets.

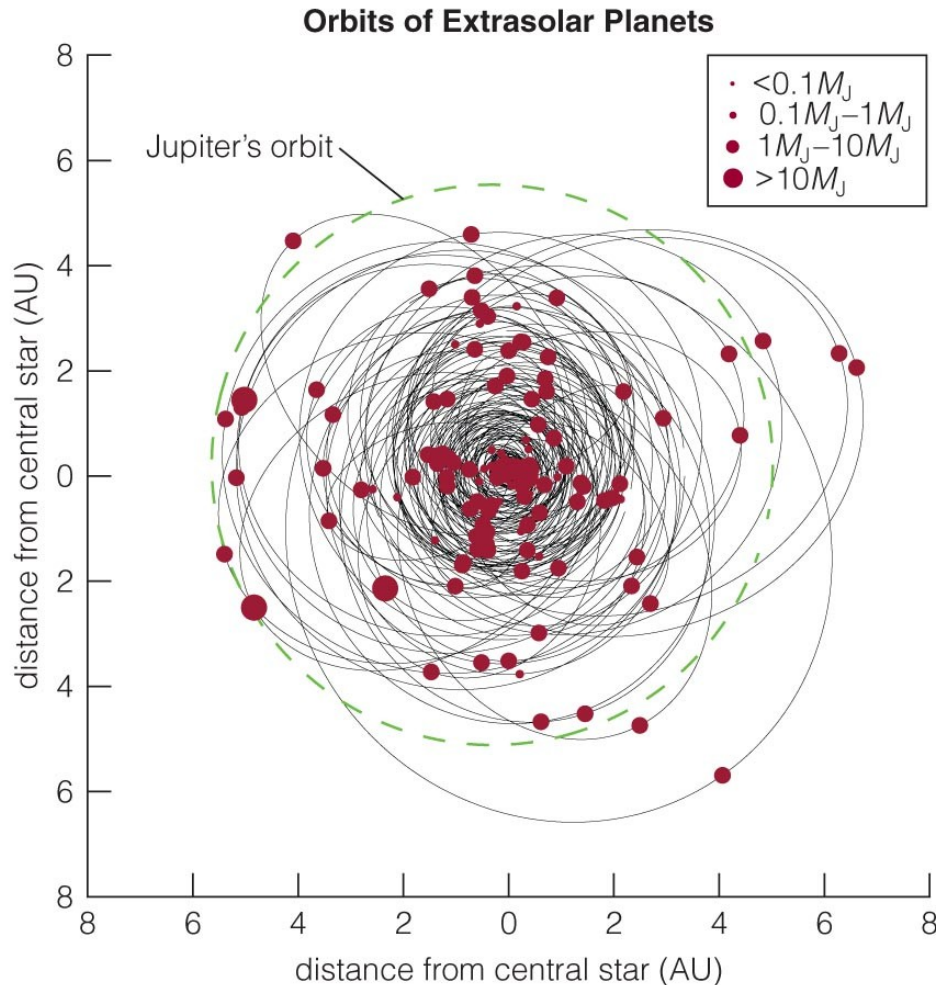
What have we learned about extrasolar planets?



Measurable Properties

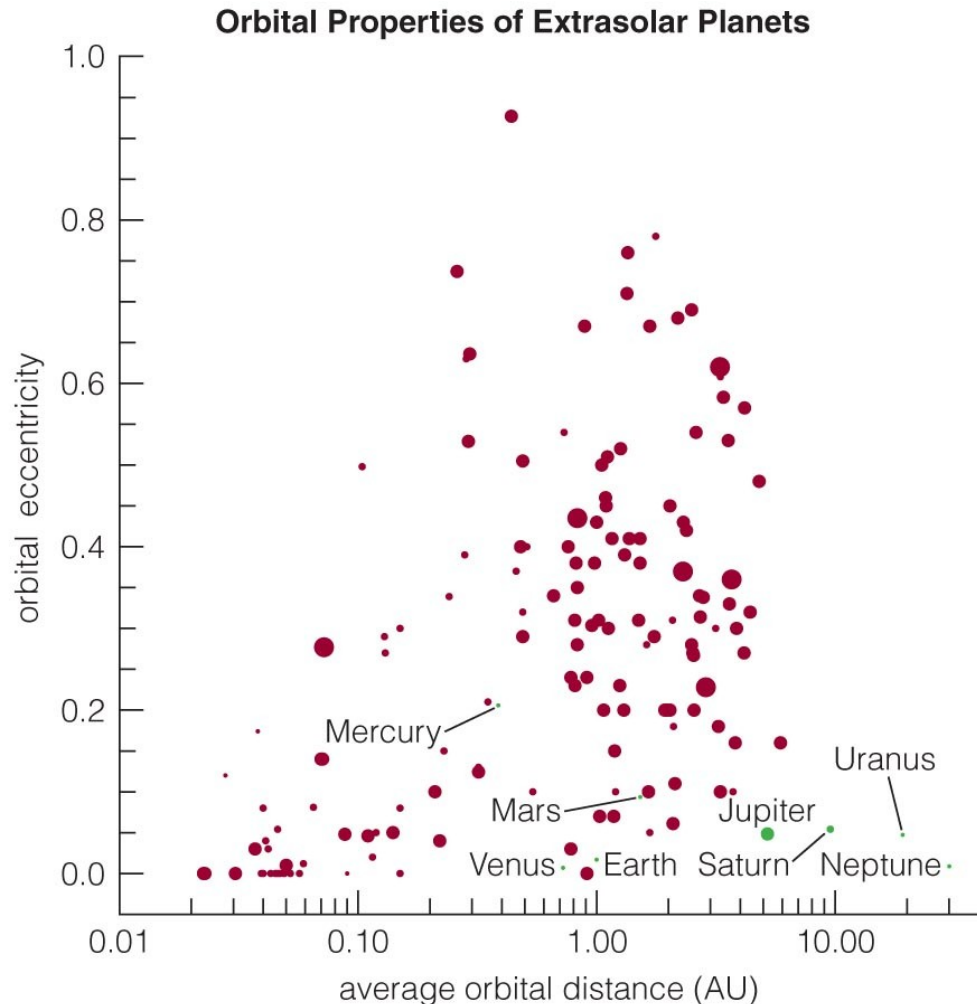
- Orbital period, distance, and Shape
- Planet mass, size, and density
- Composition

Orbits of Extrasolar Planets



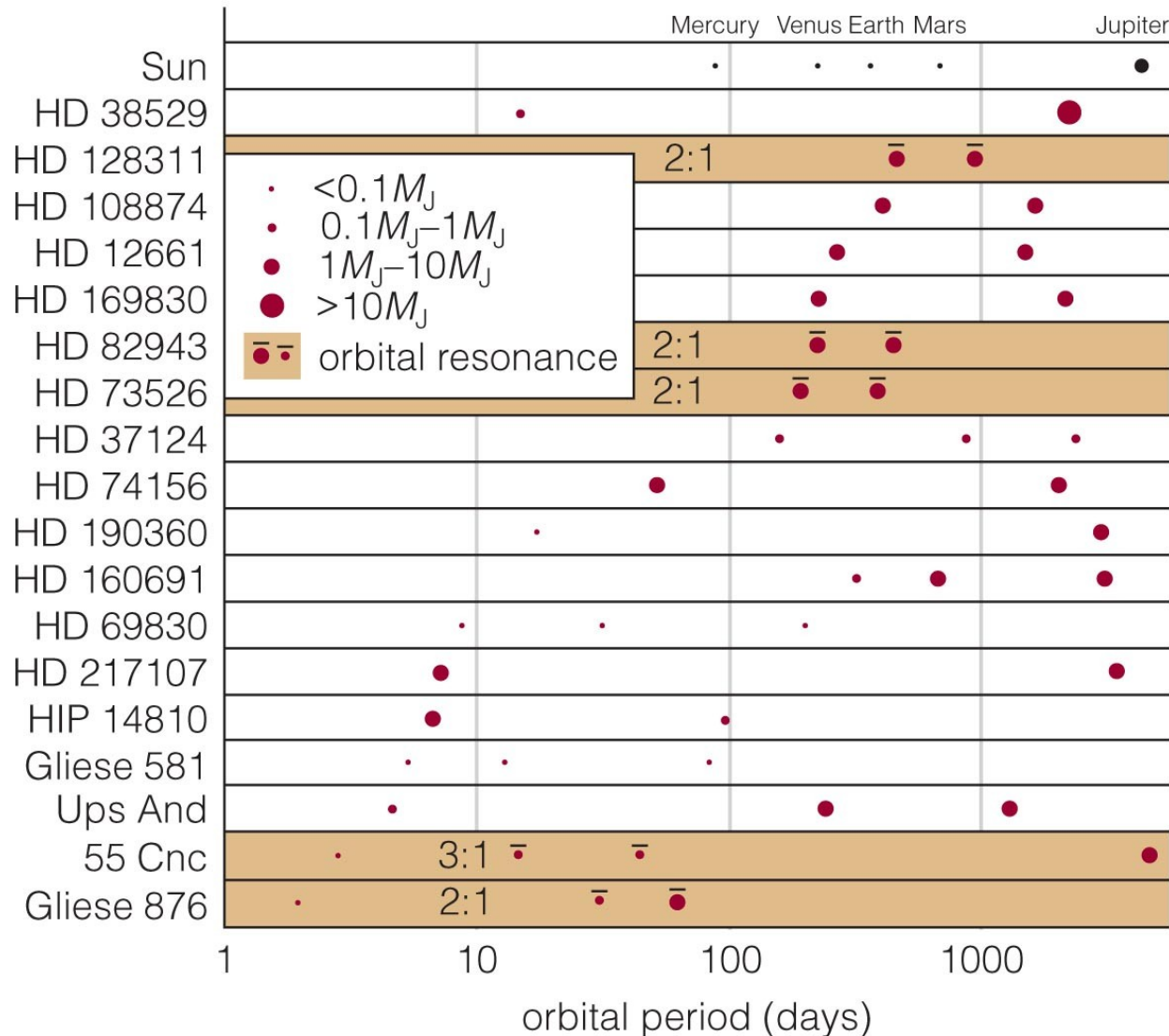
- Most of the detected planets have orbits smaller than Jupiter's.
- Planets at greater distances are harder to detect with the Doppler technique.

Orbits of Extrasolar Planets



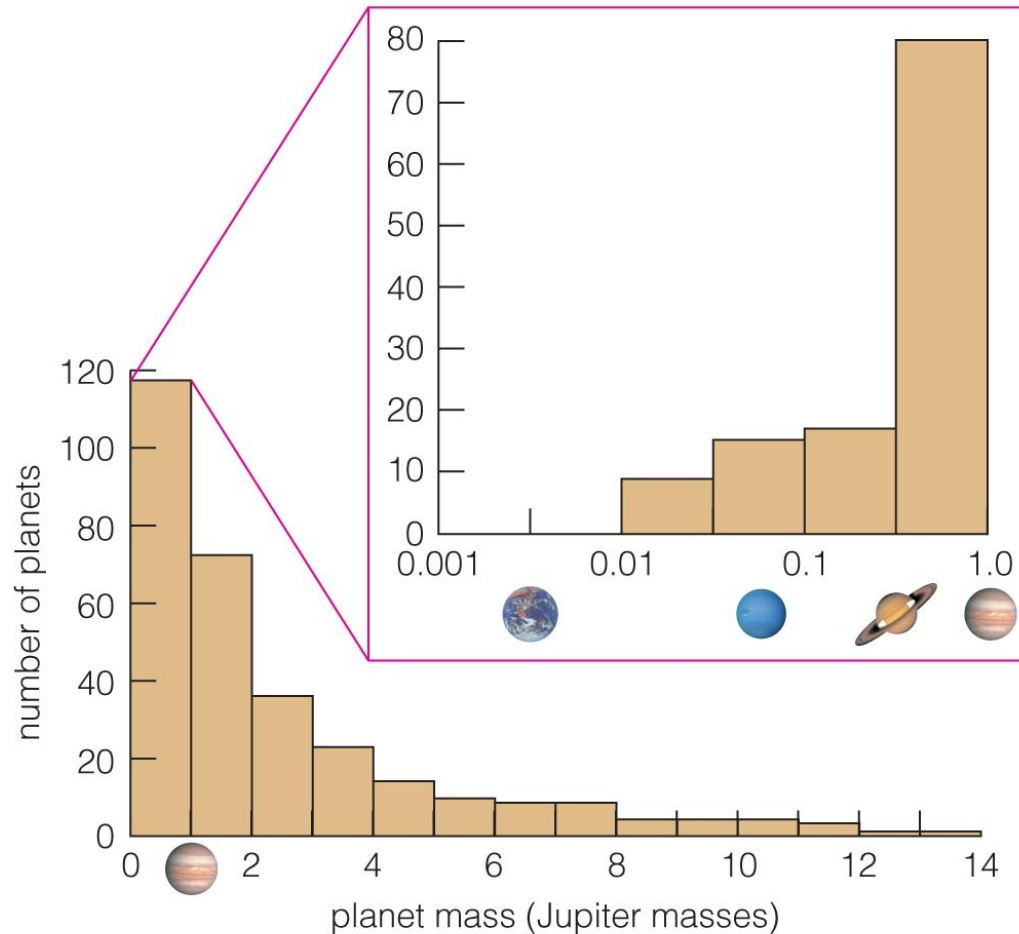
- Orbits of some extrasolar planets are much more elongated (have a greater eccentricity) than those in our solar system.

Multiple-Planet Systems



- Some stars have more than one detected planet.

Orbits of Extrasolar Planets



- Most of the detected planets have greater mass than Jupiter.
- Planets with smaller masses are harder to detect with Doppler technique.

Hot Jupiters



Jupiter

Composed primarily of hydrogen and helium
5 AU from the Sun
Orbit takes 12 Earth years
Cloudtop temperatures ≈ 130 K
Clouds of various hydrogen compounds
Radius = 1 Jupiter radius
Mass = 1 Jupiter mass
Average density = 1.33 g/cm^3
Moons, rings, magnetosphere



Hot Jupiters orbiting other stars

Composed primarily of hydrogen and helium
As close as 0.03 AU to their stars
Orbit as short as 1.2 Earth days
Cloudtop temperatures up to 1300 K
Clouds of "rock dust"
Radius up to 1.3 Jupiter radii
Mass from 0.2 to 2 Jupiter masses
Average density as low as 0.2 g/cm^3
Moons, rings, magnetospheres: unknown

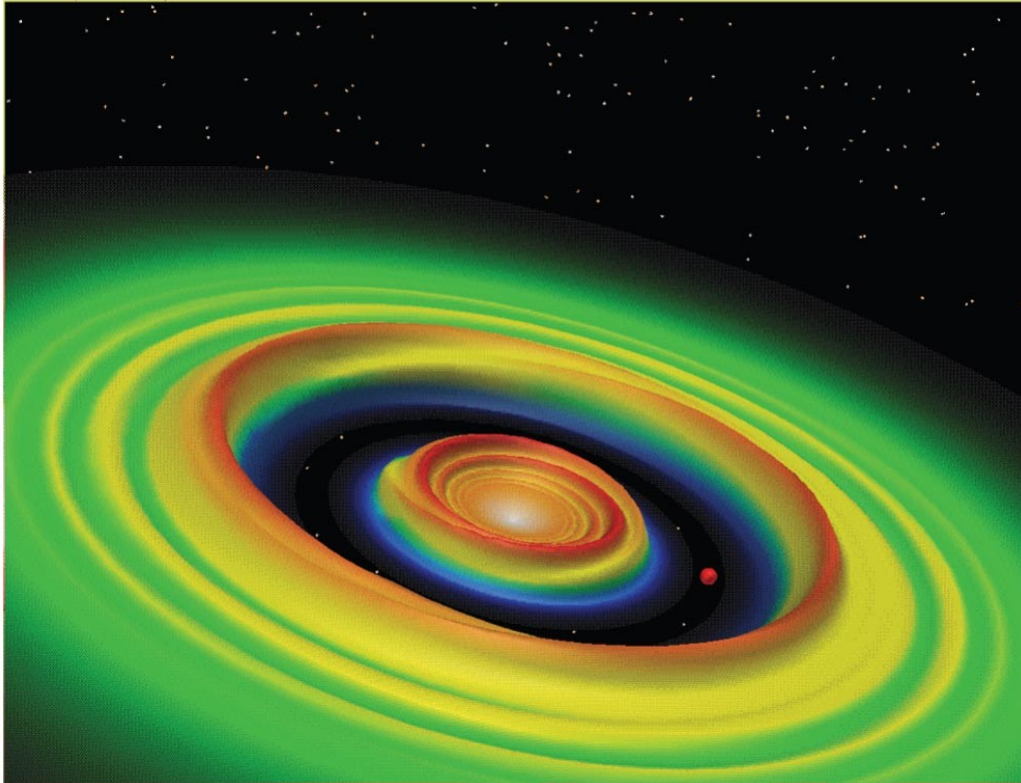
What have we learned?

- What have we learned about extrasolar planets?
 - Extrasolar planets are generally much more massive than Earth.
 - They tend to have orbital distances smaller than Jupiter's.
 - Some have highly elliptical orbits.
- How do extrasolar planets compare with planets in our solar system?
 - Some hot Jupiters have been found.

Revisiting the Nebular Theory

- The nebular theory predicts that massive Jupiter-like planets should not form inside the frost line (at $\ll 5$ AU).
- The discovery of hot Jupiters has forced reexamination of nebular theory.
- *Planetary migration* or gravitational encounters may explain hot Jupiters.

Planetary Migration



- A young planet's motion can create waves in a planet-forming disk.
- Models show that matter in these waves can tug on a planet, causing its orbit to migrate inward.

Gravitational Encounters

- Close gravitational encounters between two massive planets can eject one planet while flinging the other into a highly elliptical orbit.
- Multiple close encounters with smaller planetesimals can also cause inward migration.

Modifying the Nebular Theory

- Observations of extrasolar planets have shown that the nebular theory was incomplete.
- Effects like planetary migration and gravitational encounters might be more important than previously thought.

Planets: Common or Rare?

- One in ten stars examined so far have turned out to have planets.
- The others may still have smaller (Earth-sized) planets that current techniques cannot detect.

13.4 Finding More New Worlds

Our goals for learning:

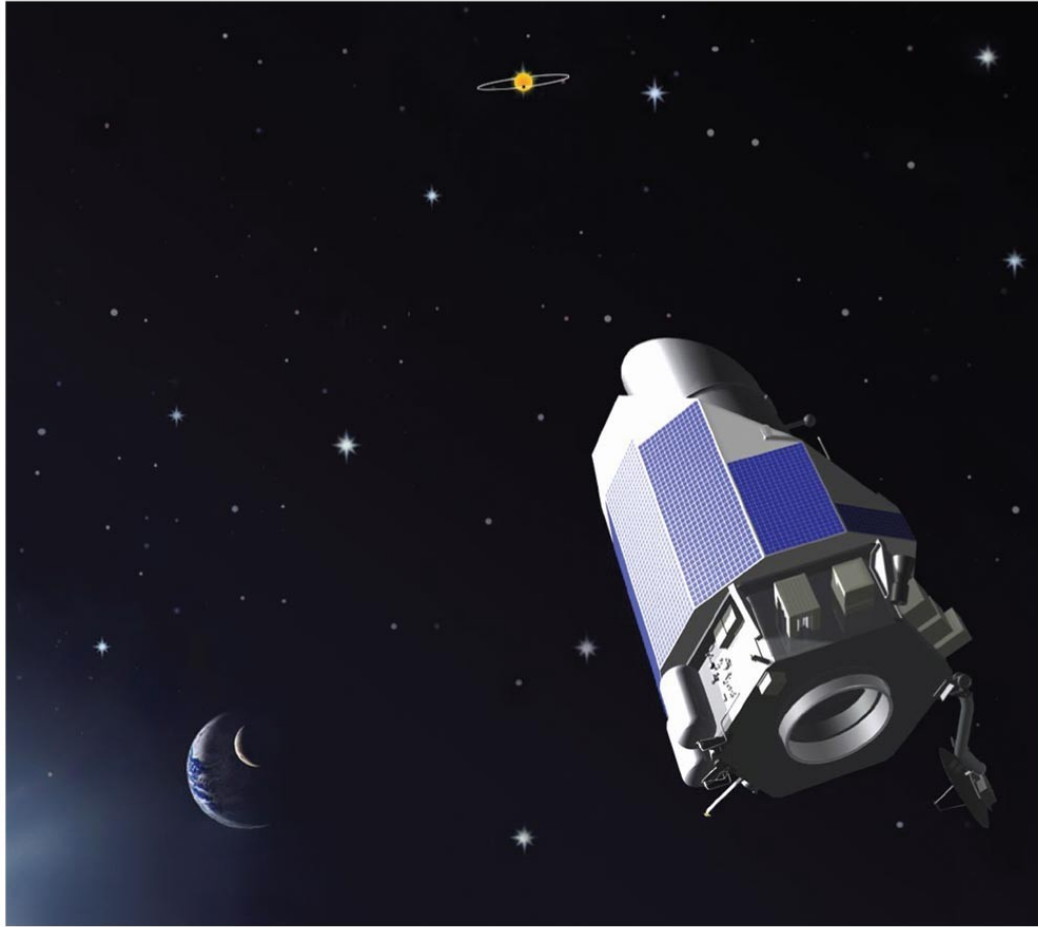
- How will we search for Earth-like planets?

How will we search for Earth-like planets?



Insert TCP 6e Figure 13.18

Transit Missions

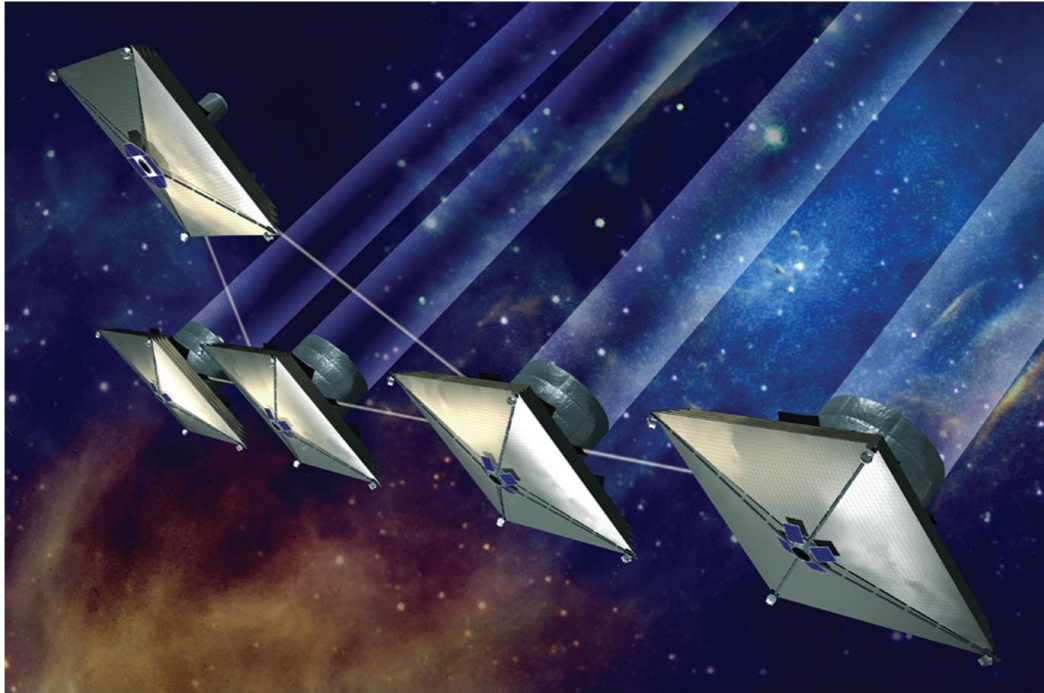


- NASA's *Kepler* mission was launched in 2008 to begin looking for transiting planets.
- It is designed to measure the 0.008% decline in brightness when an Earth-mass planet eclipses a Sun-like star.

Astrometric Missions

- *GAIA*: a European mission planned for 2011 that will use interferometry to measure precise motions of a billion stars
- *SIM*: A NASA mission that will use interferometry to measure star motions even more precisely (to 10^{-6} arcseconds)

Direct Detection



Mission concept for NASA's
Terrestrial Planet Finder (TPF)

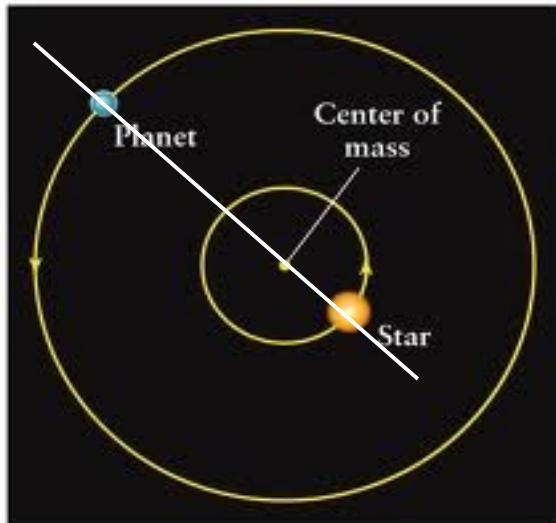
- Determining whether Earth-mass planets are really Earth-like requires direct detection.
- Missions capable of blocking enough starlight to measure the spectrum of an Earth-like planet are being planned.

What have we learned?

- How will we search for Earth-like planets?
 - Transit missions will be capable of finding Earth-like planets that cross in front of their stars.
 - Astrometric missions will be capable of measuring the “wobble” of a star caused by an orbiting Earth-like planet.
 - Missions for direct detection of an Earth-like planet will need to use special techniques (like interferometry) for blocking starlight.

p. 117 introductory astronomy lecture-tutorials

Remember : the star and the planet are always
Across from other (relative to the center of mass)



a

Also they move in opposite Direction

The wobble of the star is due to the gravitational tug from the planet. Increase the mass of the planet, increase the tug, increase the speed of the star.

You detect the wobble from the light emitted FROM THE STAR. Away the light is redshifted And the radial velocity is positive . Toward the Light is blueshifted and velocity is positive.